Basic Principles of Medicine 1
Module : Foundations to Medicine
Lecture No: 47

Title: Genetics of oncogenes

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Recommended Reading

- **Human Genetics and Genomics, Korf and Irons, 4th Edition Chapter 8 and 15**
  - **You can find the following pages and chapters on your Sakia site under:**
  - Resources/General Information/Book Access

Objectives: Genetics of Oncogenes

SOM.MK.III.BPM1.1.FTM.3.GNET.0130	Define and differentiate between a protooncogene and an oncogene. Summarize the various ways in which oncogenes may transform cells
SOM.MK.III.BPM1.1.FTM.3.GNET.0131	Compare and contrast tumor suppressor genes, DNA repair genes, and oncogenes
SOM.MK.III.BPM1.1.FTM.3.GNET.0132	Identify the different classes and functionality of proteins involved in the major cellular proliferative pathway and how mutations in which can result in cancer which includes: Growth factors (hormones), growth factor receptors, cytosolic signal transduction proteins, DNA binding transcription factors
SOM.MK.III.BPM1.1.FTM.3.GNET.0133	Identify the different types of mutation that would result in the activation of a protooncogene to an oncogene such as point mutation, nonsense mutation, chromosomal translocation, and gene amplification; provide examples for each)
SOM.MK.III.BPM1.1.FTM.3.GNET.0134	Indicate the specific chromosomal translocation in chronic myeloid leukemia and Burkitt lymphoma. Indicate the genes that are affected by the translocation in each cancer and the molecular basis of tumorigenesis in each example. Interpret a cytogenetic test that detects translocations
SOM.MK.III.BPM1.1.FTM.3.GNET.0135	Relate the pharmaceutical application of the molecular mechanisms of cancer to the development of anticancer drugs. Include the following specific examples: Herceptin in HER2+ breast cancer, Imaniteb mesylate for Ph+ chronic myeloid leukemia
SOM.MK.III.BPM1.1.FTM.3.GNET.0136	Compare and contrast the function of the following common oncogenes: Receptor Tyrosine Kinase, Ras, transcription factors
SOM.MK.III.BPM1.1.FTM.3.GNET.0137	Comprehend the progressive and cumulative nature of cancer at the genetic level using the example of FAP - Compare and contrast FAP to HNPCC (Lynch syndrome)
SOM.MK.III.BPM1.1.FTM.3.GNET.0138	Compare and contrast the function of the following common tumor suppressor genes: RB, p53, Wilms tumor gene, APC

Required Reading and Objectives

After 3 lectures on Cancer Genetics: Genetics of Oncogenes; Genetics of Tumor Suppressor Genes & DNA Repair.

Apoptosis and Cancer

1. Lippincott Illustrated Reviews: Cell and Molecular Biology, Chapter 22

[Abnormal Cell Growth | Lippincott® Illustrated Reviews: Cell and Molecular Biology, 3e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)]

The following sections only:
Genes and Cancer: Abnormal cell growth: p53 – the guardian of the genome

2. Lippincott Illustrated Reviews: Cell and Molecular Biology, Chapter 23

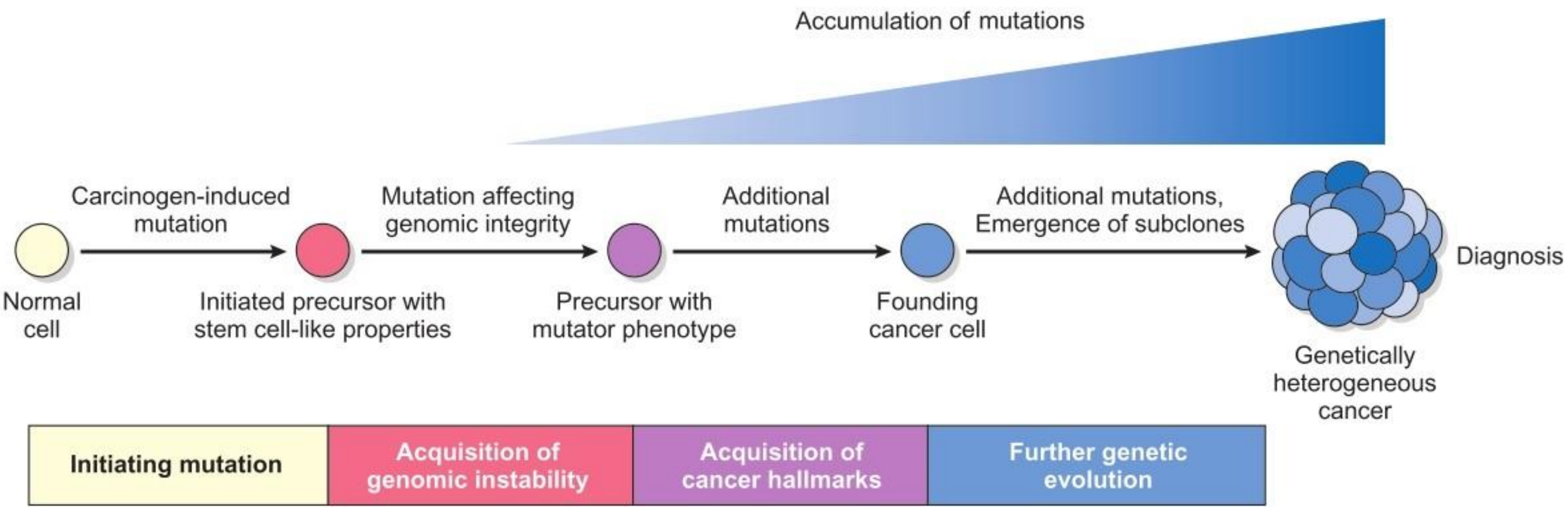
[Cell Death | Lippincott® Illustrated Reviews: Cell and Molecular Biology, 3e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)]

The following sections only:
Cell Death: Initiation of apoptosis Including figure 23.7
Bcl-2 family including figures 23.13 and 23.14
Apoptosis in disease
Cancer

SOM.MK.III.BPM1.1.FTM.3.GNET.0150	Discuss the central role of p53 in regulation of cell proliferation, cell survival, and apoptosis
SOM.MK.III.BPM1.1.FTM.3.GNET.0151	Discuss the role of BAX and BCL-2 in regulating apoptosis: -BAX as pro-apoptotic, -BCL-2 as anti-apoptotic
SOM.MK.III.BPM1.1.FTM.3.GNET.0152	Indicate how cytochrome c release from mitochondria may initiate apoptosis
SOM.MK.III.BPM1.1.FTM.3.GNET.0153	Summarize the extrinsic, intrinsic, and perforin pathway of apoptosis with respect to cancer.

What is Cancer?
Why does it happen?

- Cancer is a term used to describe a group of disorders characterized by aberrant cell growth.
- This is caused by a series of acquired mutations affecting a single cell and its clonal progeny.
- The mutations usually target genes that regulate proliferation, differentiation, and survival of cells.



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Darwinian Microevolution

- All cancer cells originate from a single cell.
- Accumulation of mutations with each successive proliferation.
- With each cumulative mutational event the mutated cells gain a growth advantage.

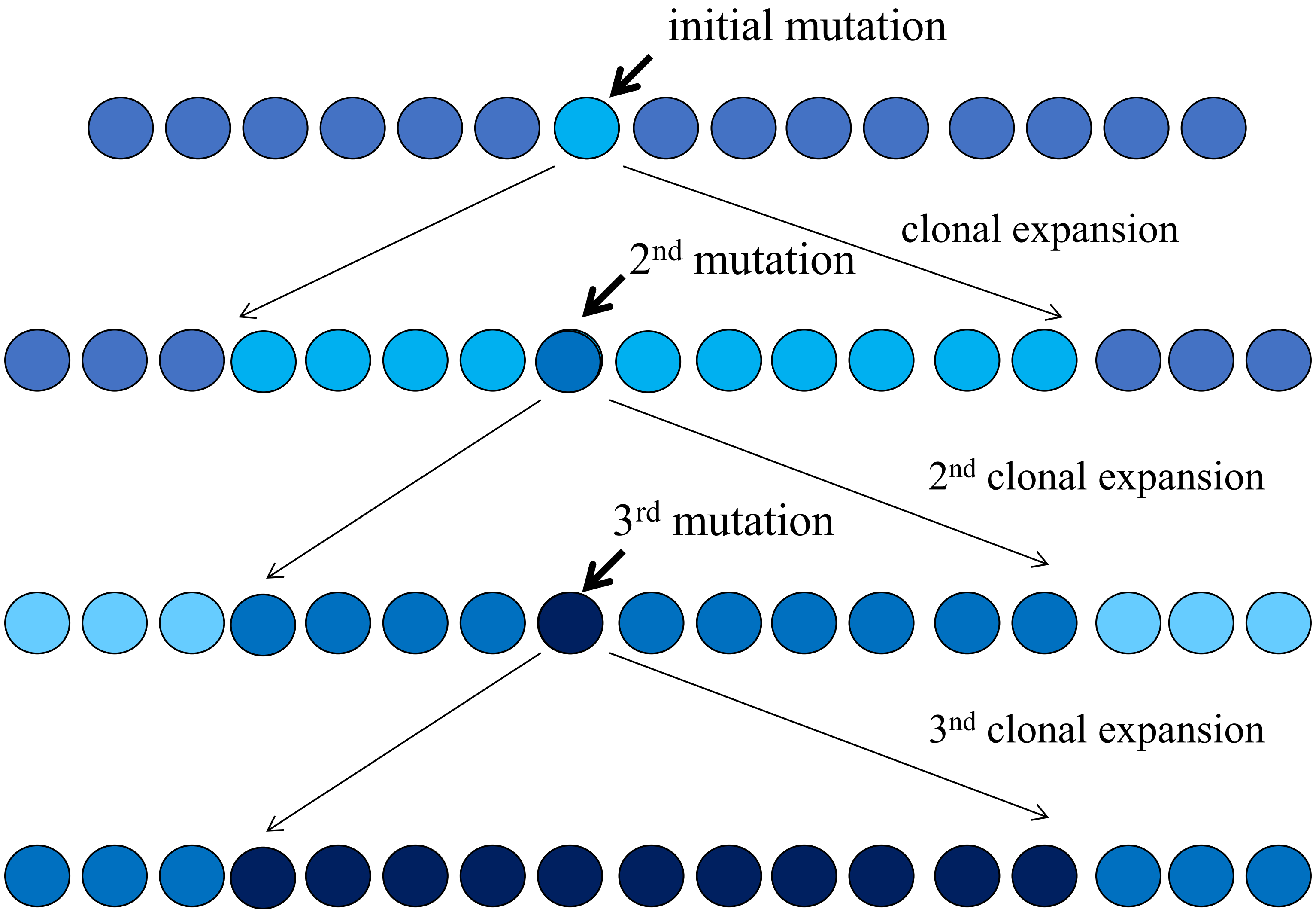
Tumor Progression

Normal → Hyperplastic → Dysplastic → Neoplastic → Metastatic

The property of **progressive aggressiveness** is selected for in cancer.

Tumor progression results from **waves of mutation** followed by **clonal expansion**.

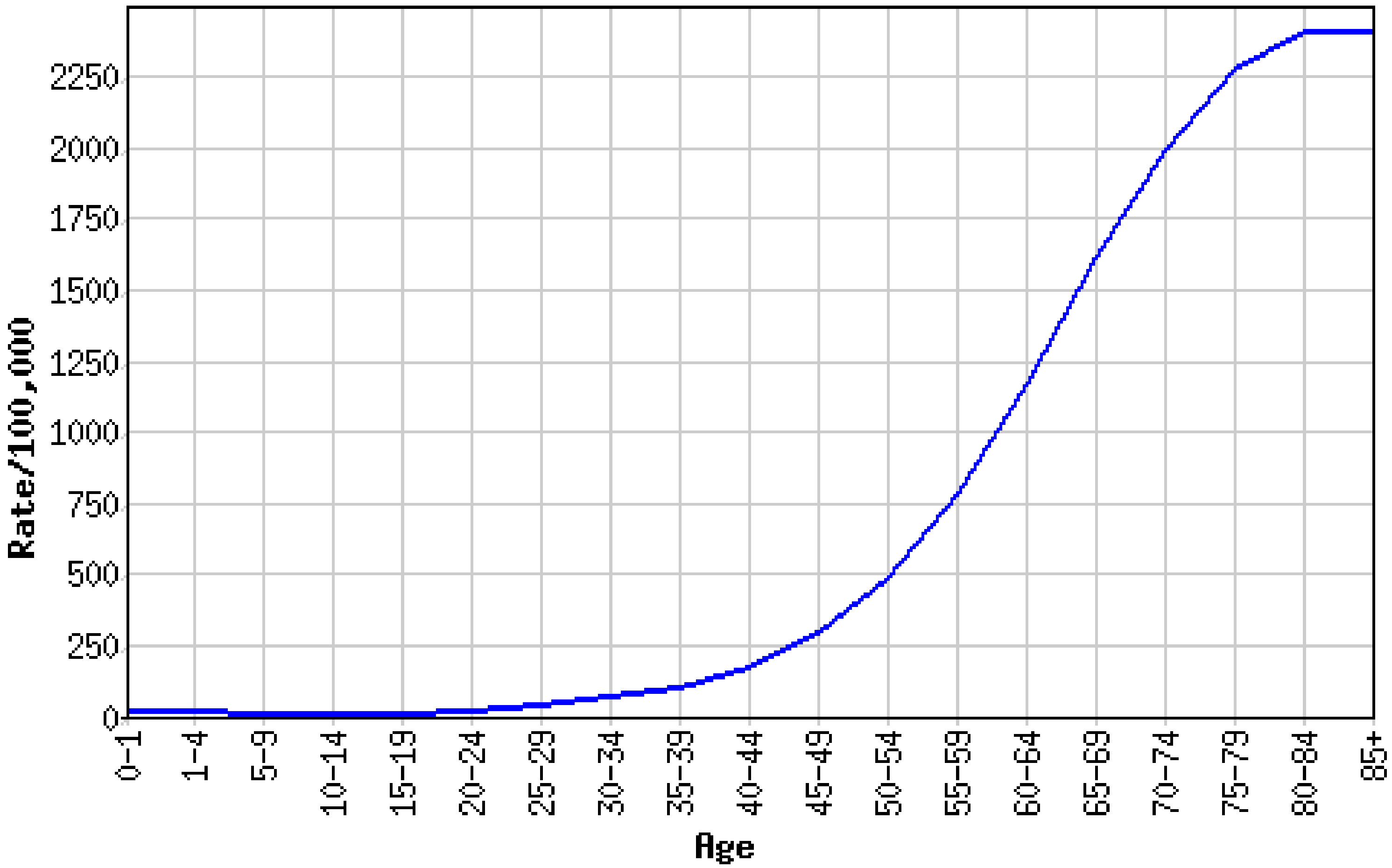
Cancer results from Darwinian evolution on a microscopic scale



Tumor heterogeneity

Age dependency of cancer (all types combined)

Most cancers occur with the 5th to 6th power of age, suggesting that they result because of multiple independent events:



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Evolution towards cancer

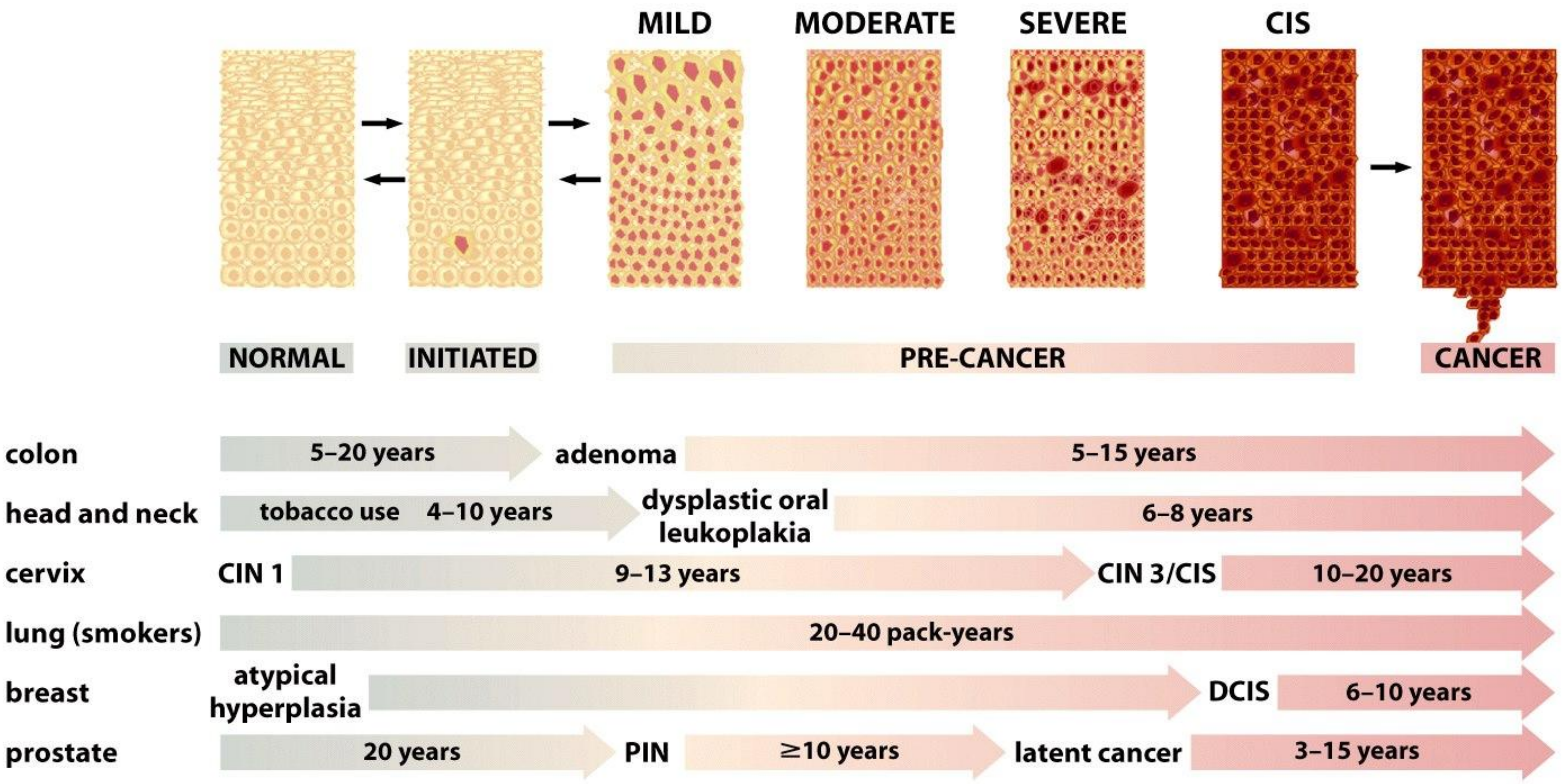

Figure 11.7 *The Biology of Cancer* (© Garland Science 2007)

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Clonal model for cancer

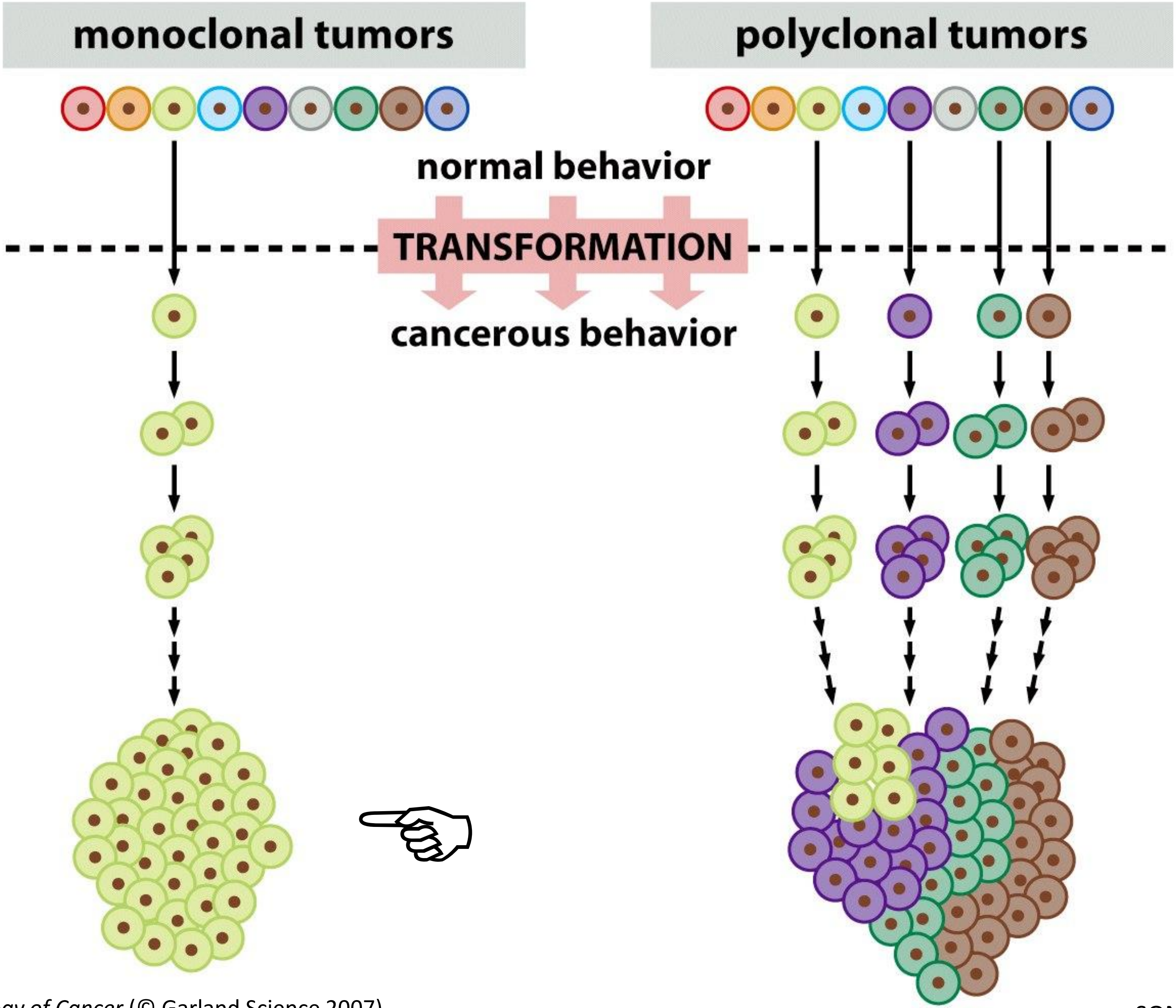
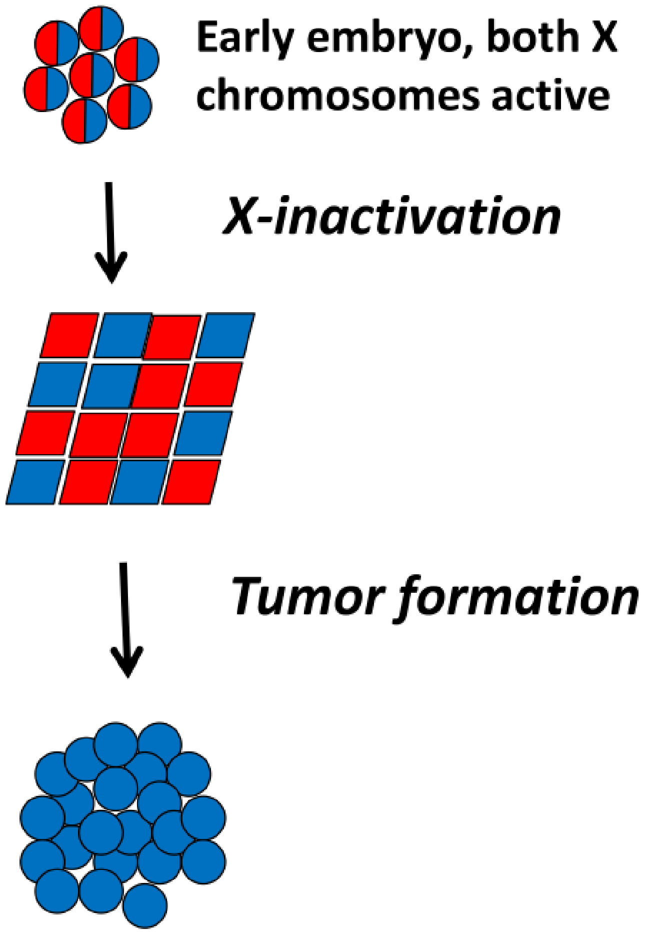


Figure 2.17 *The Biology of Cancer* (© Garland Science 2007)

Evidence of Monoclonality

1. **Examination of X-inactivation pattern in cancers**

- Tissues are mosaic for X-inactivation
- BUT, cells from cancer tissue have the same copy of the inactivated X-chromosome
- Strongly suggests that they are derived from a single cell



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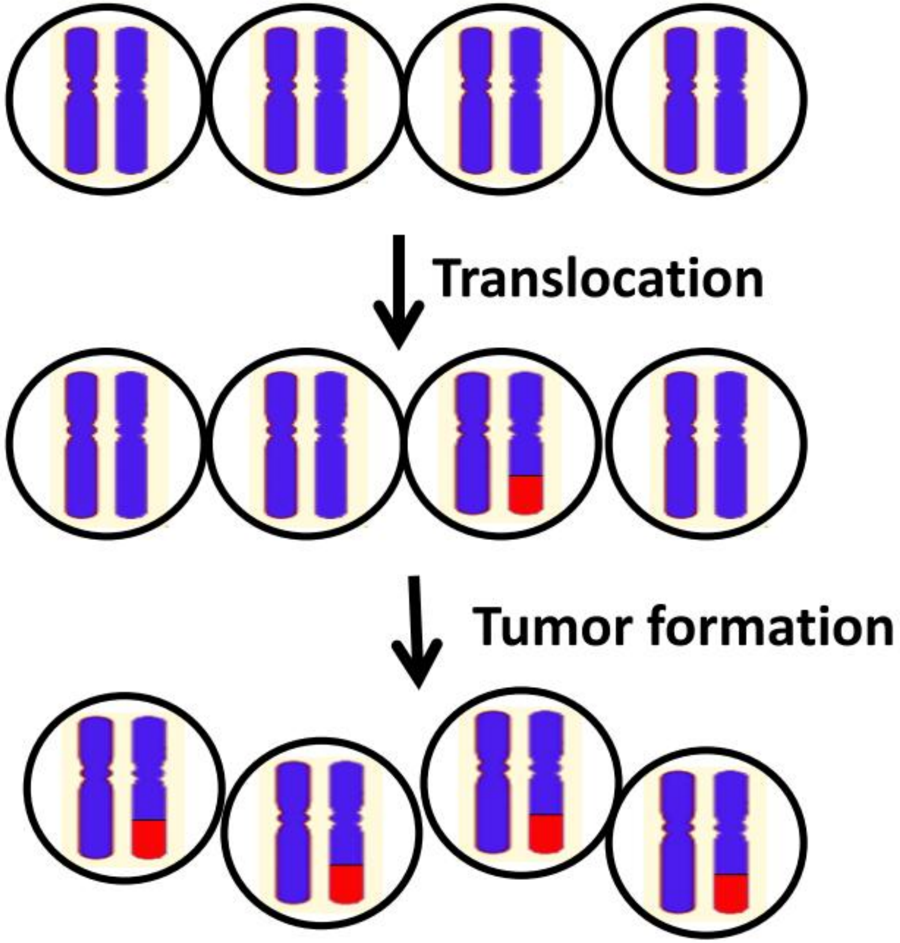
Evidence of Monoclonality

2. Chromosomal abnormalities of cancers

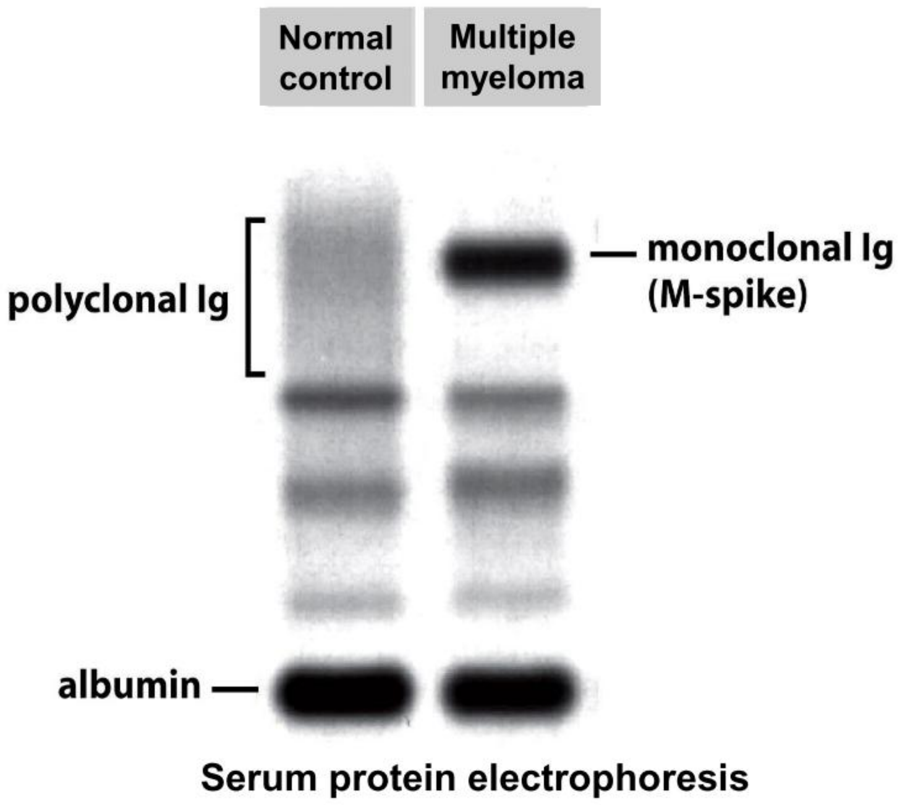
- Chromosomal aberration occurs
- Cancer develops
- All cells in the tumor contain the genetic aberration



Evidence of Monoclonality

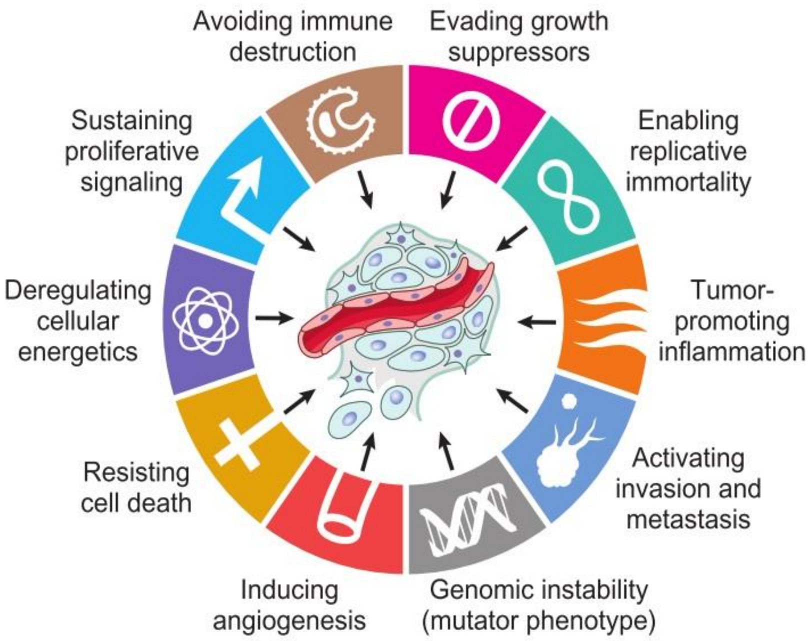
3. Multiple Myeloma produce a monoclonal Immunoglobulin

- Multiple myeloma is a malignancy of B-cell (precursors of antibody-producing plasma cells).
- All Myeloma cells in a patient produce the same antibody molecule.
- Also evidence for monoclonality of cancers.



Fundamental Changes for Cancer Formation

1. Proliferation without external stimuli because of oncogenes activation.
2. Nonresponsive to growth inhibition because of tumor suppressor genes inactivation.
3. Metabolic switch to aerobic glycolysis for rapid growth.
4. Resistance to programmed cell death.
5. Unrestricted proliferative capacity.
6. Induce angiogenesis.
7. Loss of cell to cell adhesion and contact inhibition of growth allowing invasion and metastasis.
8. Evasion of the immune system.



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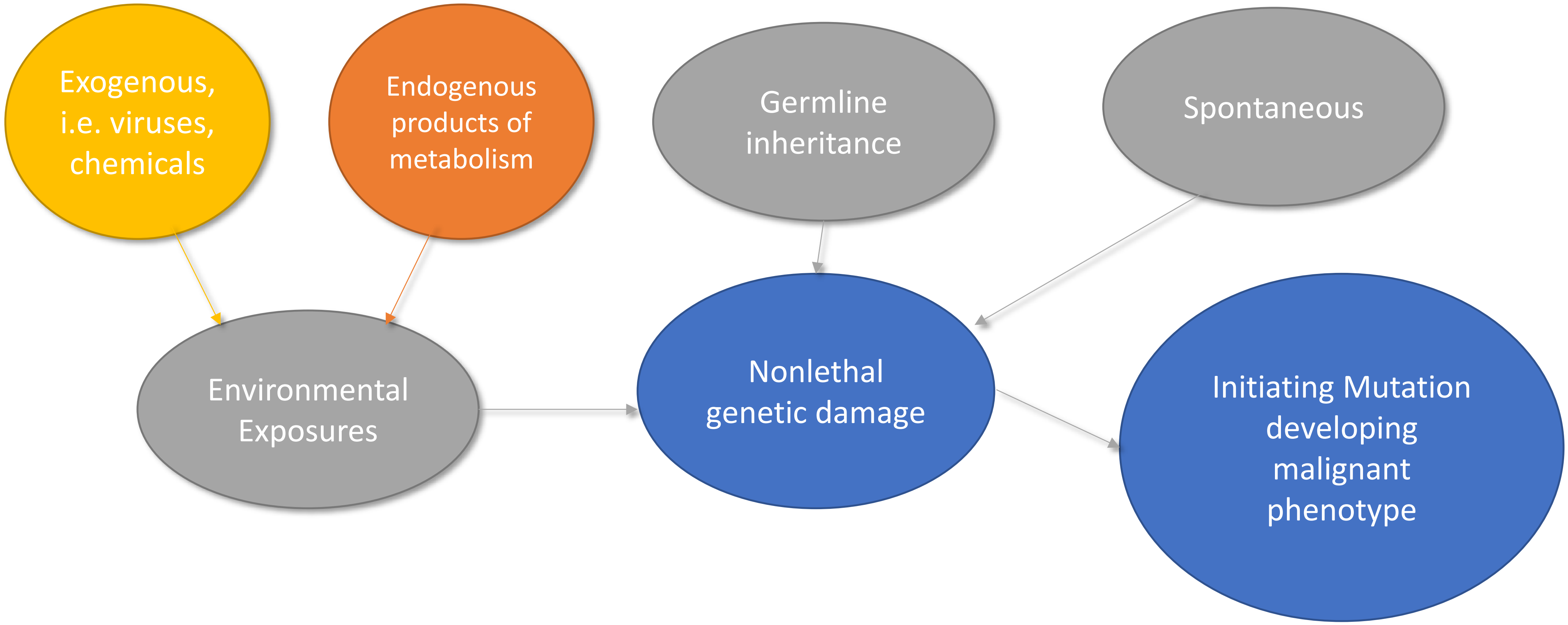
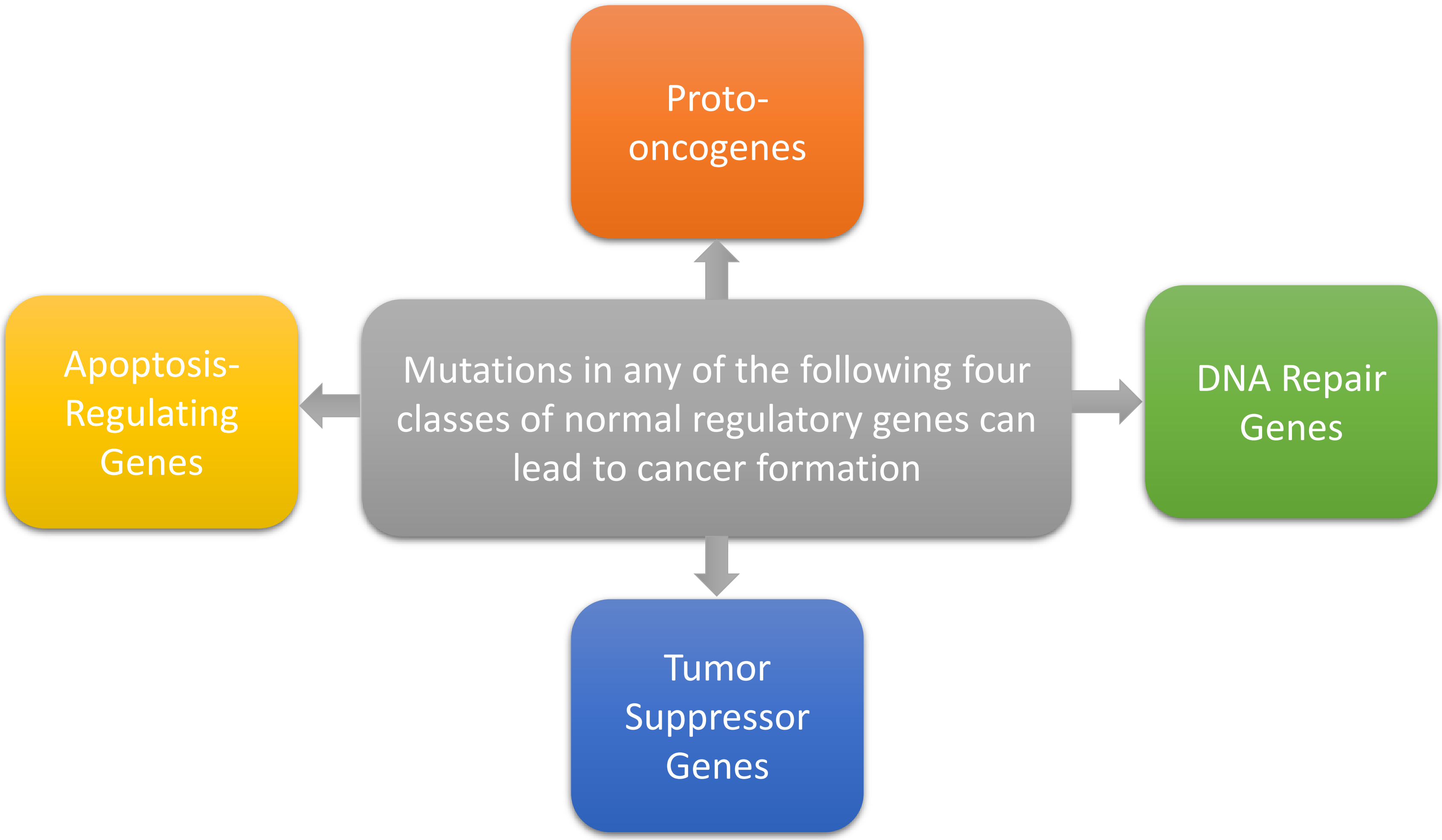
How do the mutations take place?



Which genes are targeted by cancer-causing mutations?



Which genes are targeted by cancer-causing mutations?

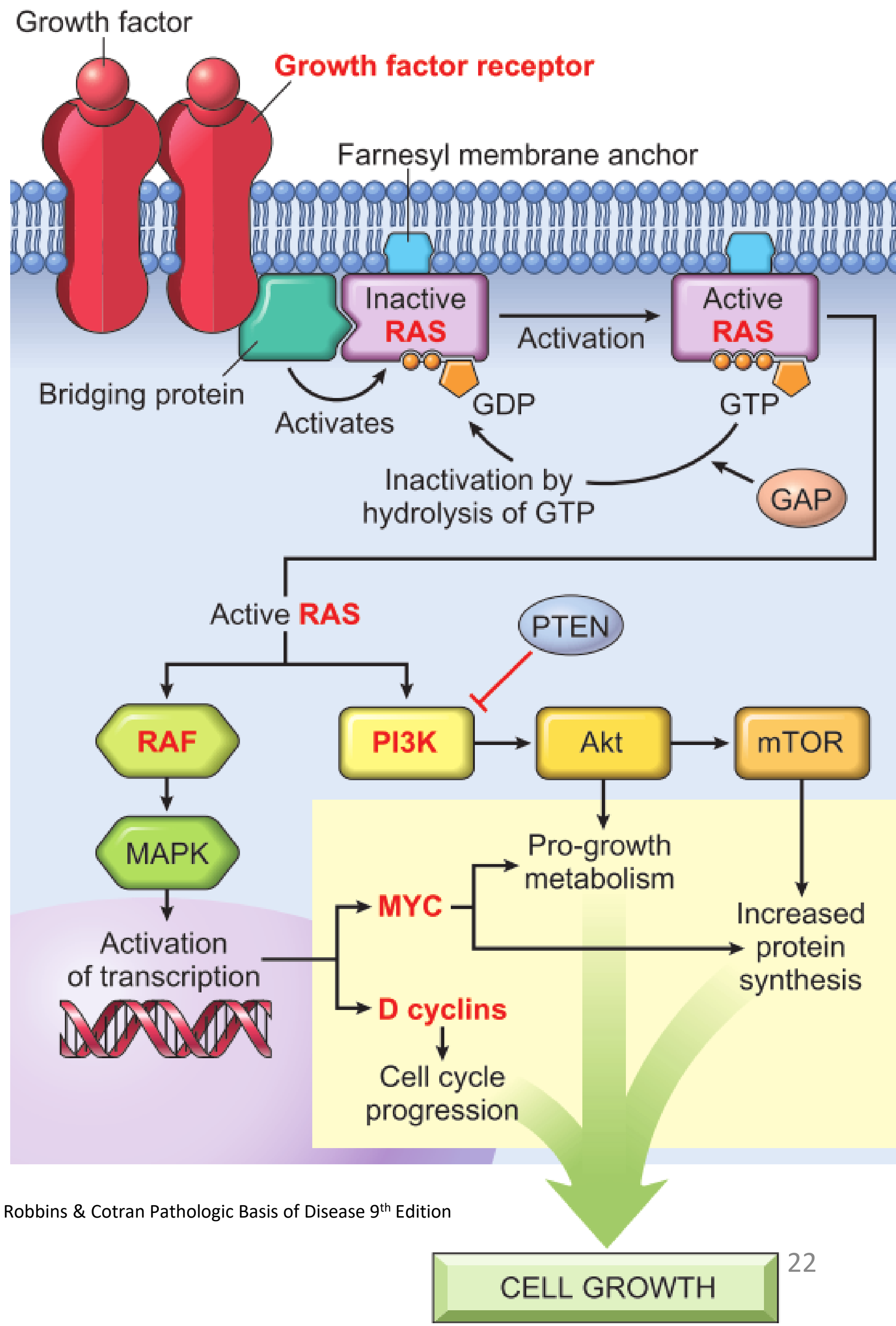
1. Proto-oncogenes:
   - Normally they encode proteins that promote cell growth
   - Mutations activating proto-oncogenes can result in one of two things:
     - Excessive increase in one or more normal function
     - Create a completely new function on the affected gene product (Gain of Function)
   - Usually a single copy mutation is sufficient to cause cancer despite the presence of a normal copy (Oncogenes are dominant)

Which genes are targeted by cancer-causing mutations?

2. Tumor Suppressor Genes:
    - Usually produces genes that inhibit cell cycle and preventing proliferation
    - Mutations usually result in a Loss of Function
    - Both alleles must be affected before malignant transformation can take place (Tumor Suppressor Genes behave in a recessive manner)
3. Apoptosis-Regulating Genes:
    - Mutations that can cause either
        - Gain of function in genes that suppress apoptosis (anti-apoptotic)
        - Loss of function in genes that promote apoptosis (pro-apoptotic)
4. DNA Repair Genes:
    - Mutations in these genes impair the cell's ability to recognize and repair nonlethal genetic damage

Major Cellular Proliferative Pathway (MAP Kinase Pathway) Under Normal Physiologic Condition

1. Growth factor presence outside the cell
2. Binding of the growth factor to its specific growth factor receptor
3. Receptor autophosphorylation
4. Exchange of GTP for GDP on RAS
5. RAS binds to RAF and begins a phosphorylation cascade
6. Upregulation and downregulation transcription of different genes favoring cell survival and proliferation



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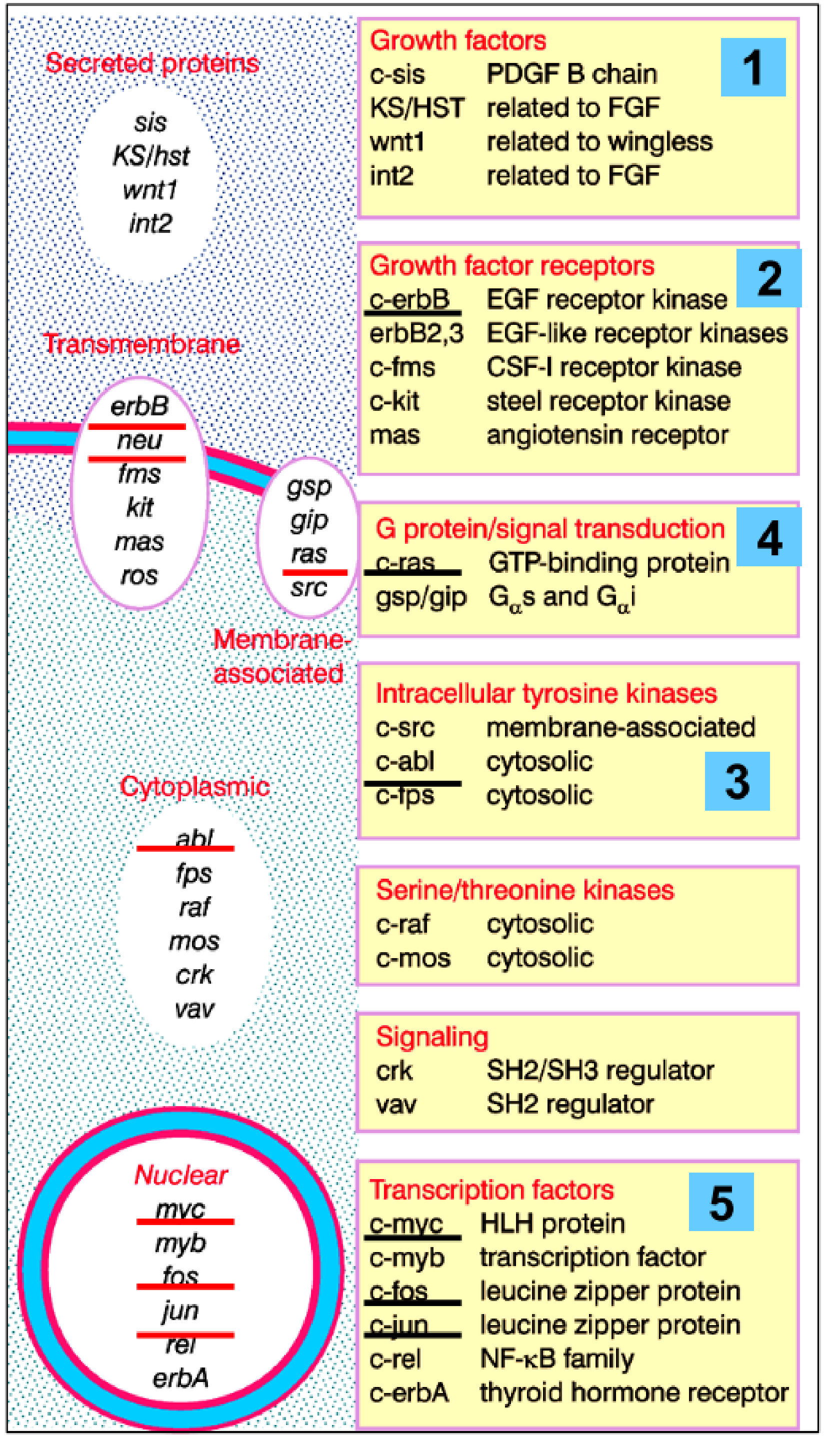
Oncogenes in the MAP Kinase Pathway

- Proto-oncogenes encode for proteins that promote cell growth and proliferation under the presence of normal growth-promoting signals.
- Mutations in the proto-oncogenes inactivate their internal regulatory elements resulting in oncogenes that can promote cell growth and proliferation in the absence of a growth-promoting signal (said to be Constitutively active).
- Mutations can occur at any of the five proteins along the signal transduction pathway. (Numbers labeled in the table on the right)

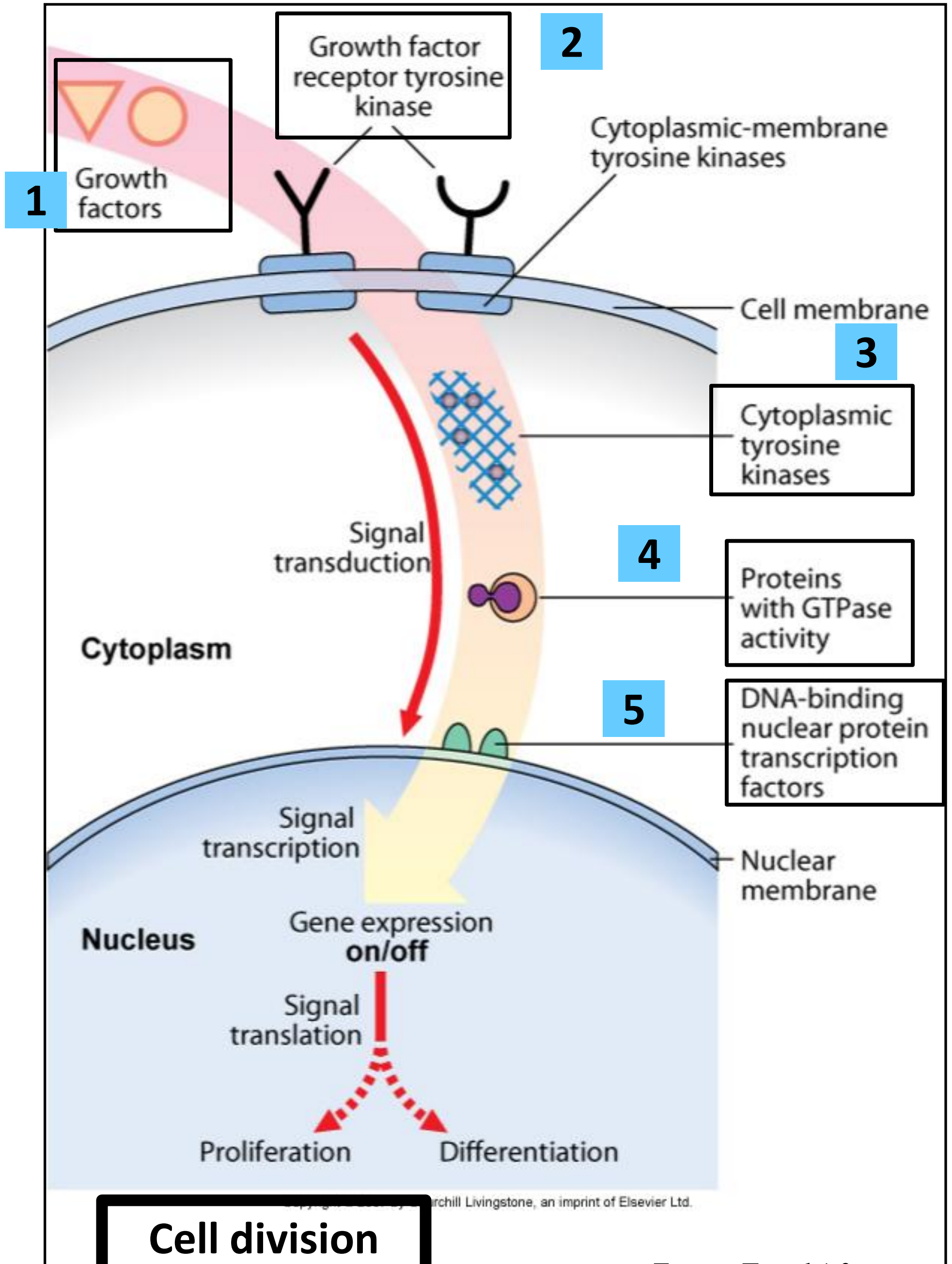


Oncogenes in the Major Cellular Proliferative Pathway (MAP kinase pathway)

Mutations in any of the five components of the Major Cellular Proliferative Pathway (MAP Kinase Pathway) results in oncogene formation:

1) Growth factors: which initiate the pathway through binding growth factor receptors.
2) Growth factors receptors: which are tyrosine kinase receptors. (Example: Her-2 receptor)
3) Intracellular/Cytoplasmic tyrosine kinases. (Example: Abl of bcr-abl).
4) Proteins with GTPase activity that are signal transducers that will start a signaling cascade that ends up activating transcription factors (Example: Ras).
5) Transcription factors: DNA-binding proteins which in turn activate gene expression to drive DNA replication and cell division. (Example: Myc).

https://www.youtube.com/watch?v=i1f2RbogiDw
https://www.youtube.com/watch?v=NlFDea3ig7A

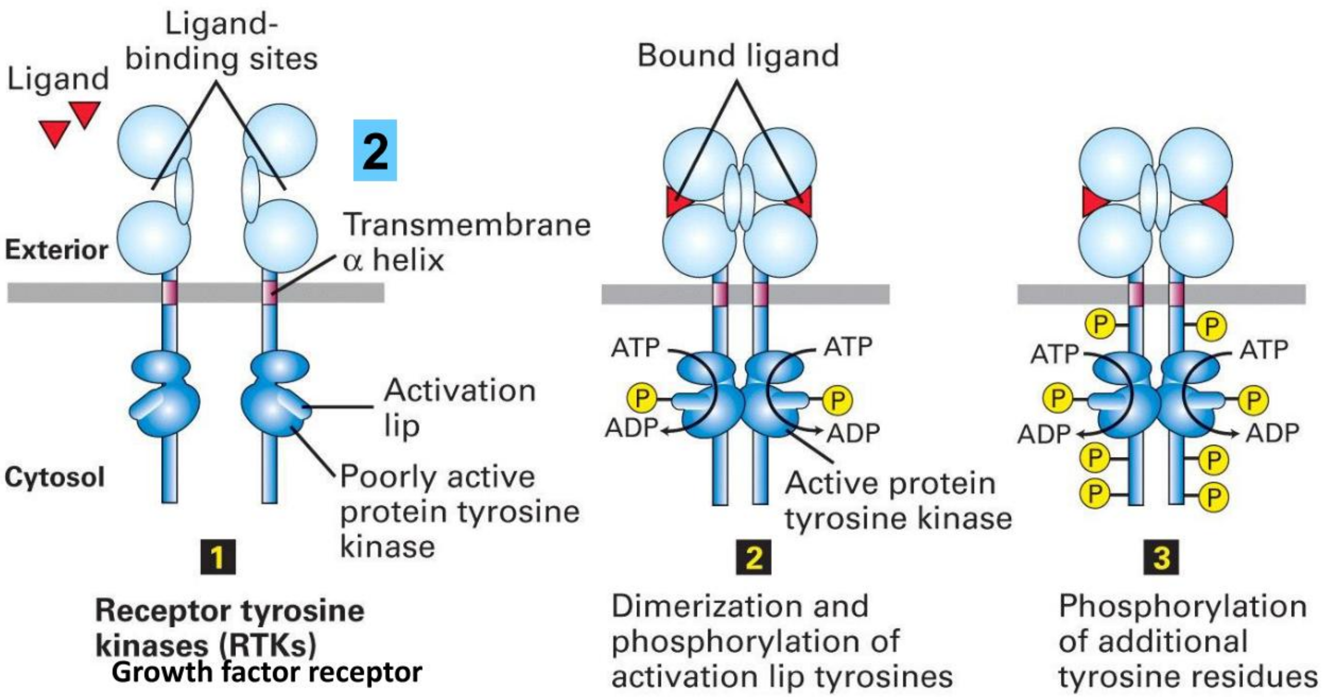


Emery Fig. 14.3

Case 1 - Breast Carcinoma

- A 42-year-old woman comes to the physician because she felt a growing mass in her left breast that she first noticed 6 months ago. Physical examination shows a non-tender, firm mass in the left breast that is approximately 2-cm in size. Her mammogram and ultrasound confirm the mass and a biopsy is taken. The biopsy shows a poorly differentiated invasive ductal carcinoma of the breast with HER2 3+ by immunohistochemistry. Based on the pathology and imaging results she is diagnosed with HER2-positive breast cancer. She is started on a neoadjuvant chemotherapy that includes Trastuzumab, a monoclonal antibody to HER2 receptors.

Growth Factor (Tyrosine Kinase) Receptor Activation



- Growth factor binds to receptor tyrosine kinase

↓

- Activates the receptor transiently

↓

- Conformational change to active dimeric state

↓

- Autophosphorylates tyrosine residues

How is Growth Factor Receptor Mutation involved in Breast Carcinomas?

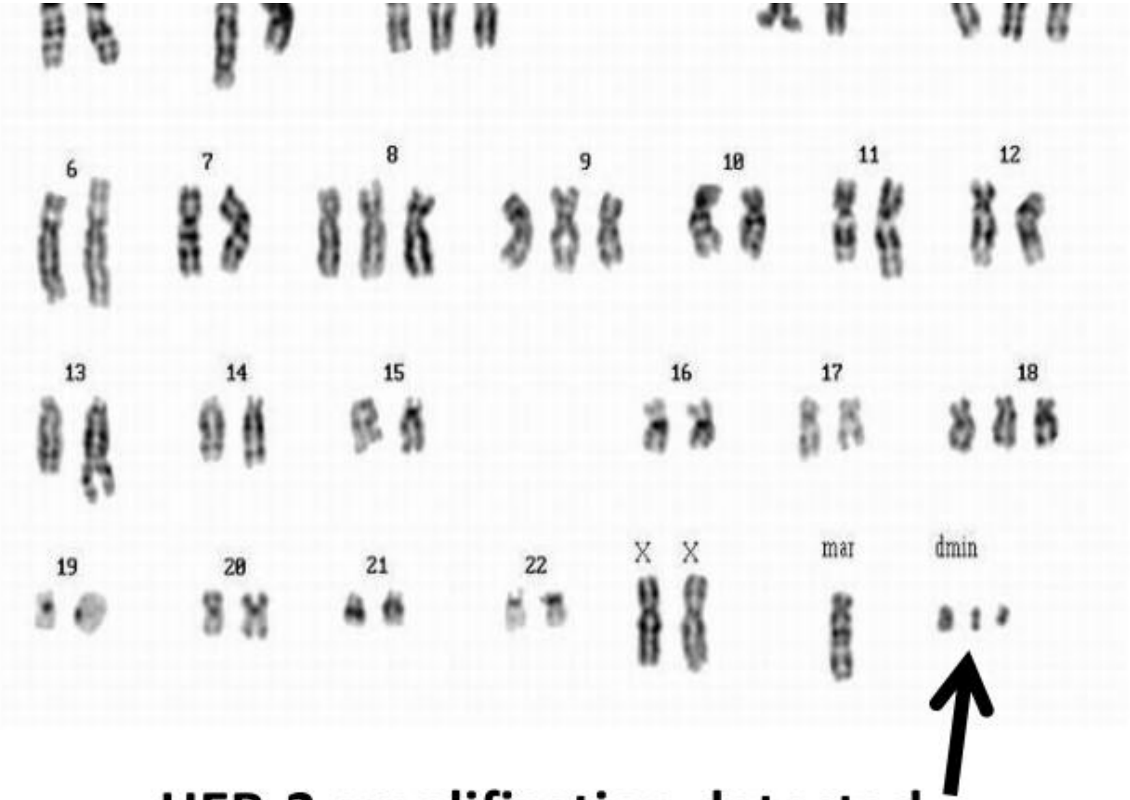
- *ERBB2* gene encodes HER2, which is a member of the receptor tyrosine kinase family.

- *ERBB2* gene is amplified in approximately 30% of breast carcinomas leading to overexpression of the HER2 receptor and that in turn will lead to a constitutively active tyrosine kinase.

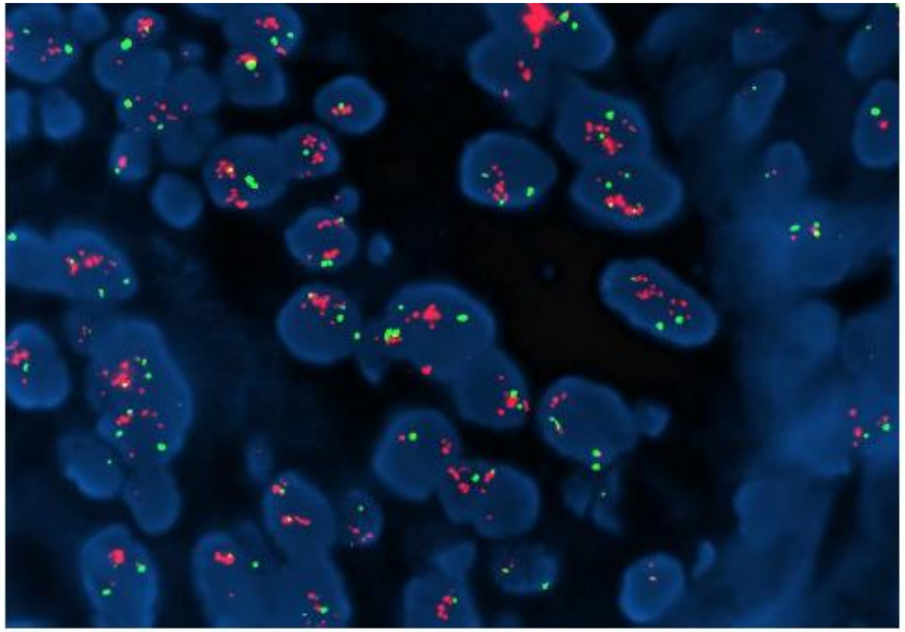
So How Are Oncogenes Activated by Gene Amplification?

- Oncogenes are activated by amplification in one of two methods:
  - Double minutes (Extrachromosomal Elements)
  - Homogeneously staining regions (hsrs): Attached to the chromosome
- In both, hundreds of copies of the oncogene is present, resulting in production of excessive unregulated amounts of the oncoprotein and a sustained stimulus to divide
- Double minutes are extrachromosomal fragments of DNA containing an amplified oncogene (hundreds of copies)
- HSRS are amplified oncogenes attached to the chromosomes

HER-2 Overexpression in Sporadic Breast Carcinoma



HER-2 amplification detected as double minutes



HER-2 amplification detected by FISH probes

Targeted Molecular Therapy in Breast Carcinoma

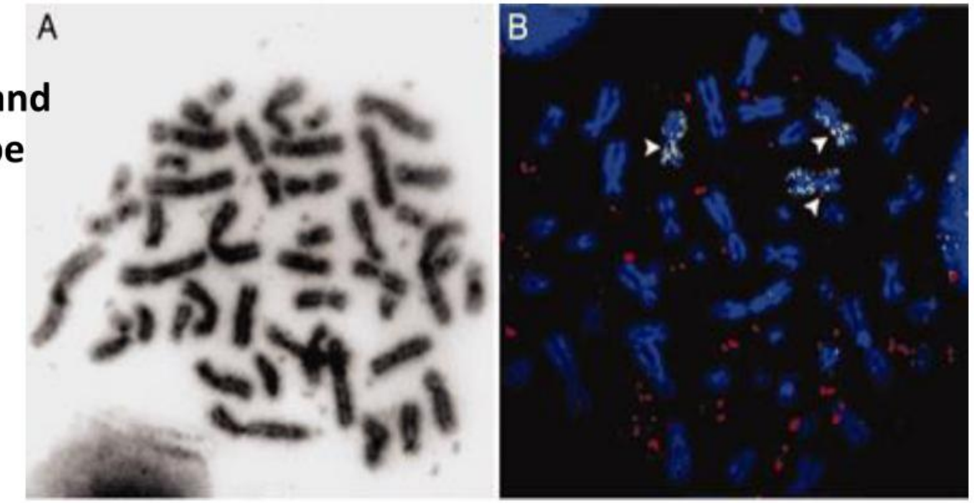
- Herceptin (Trastuzumab) is a monoclonal antibody against HER2 receptor blocking its activity.

- Patients who have breast carcinomas with *ERBB2* amplification and HER2 overexpression respond to Herceptin.

- Herceptin will cause tumor size regression, stops tumor growth, and induces apoptosis of the tumor cells.

Oncogene Amplification: Double minutes

From G-band karyotype

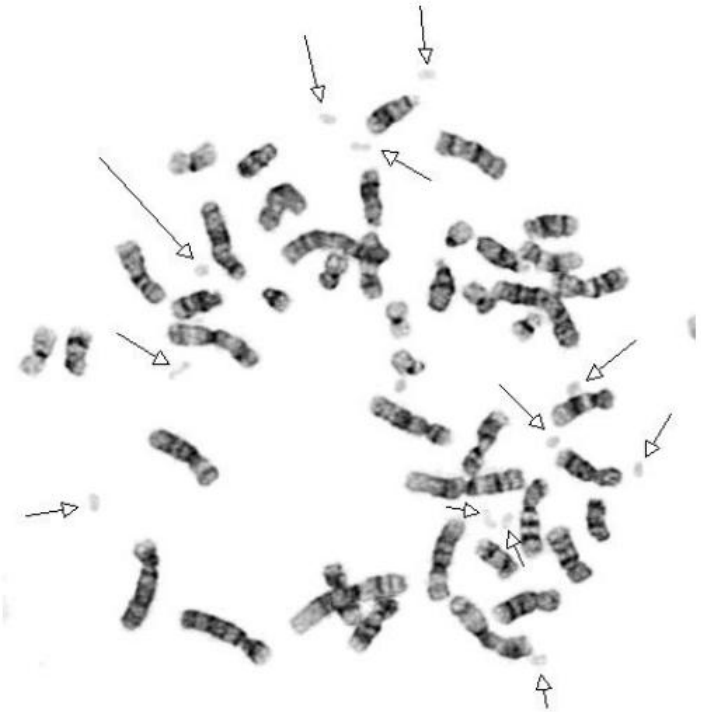


Visualized with FISH probes

Vogt et al. 2004 PNAS *101: 11368*

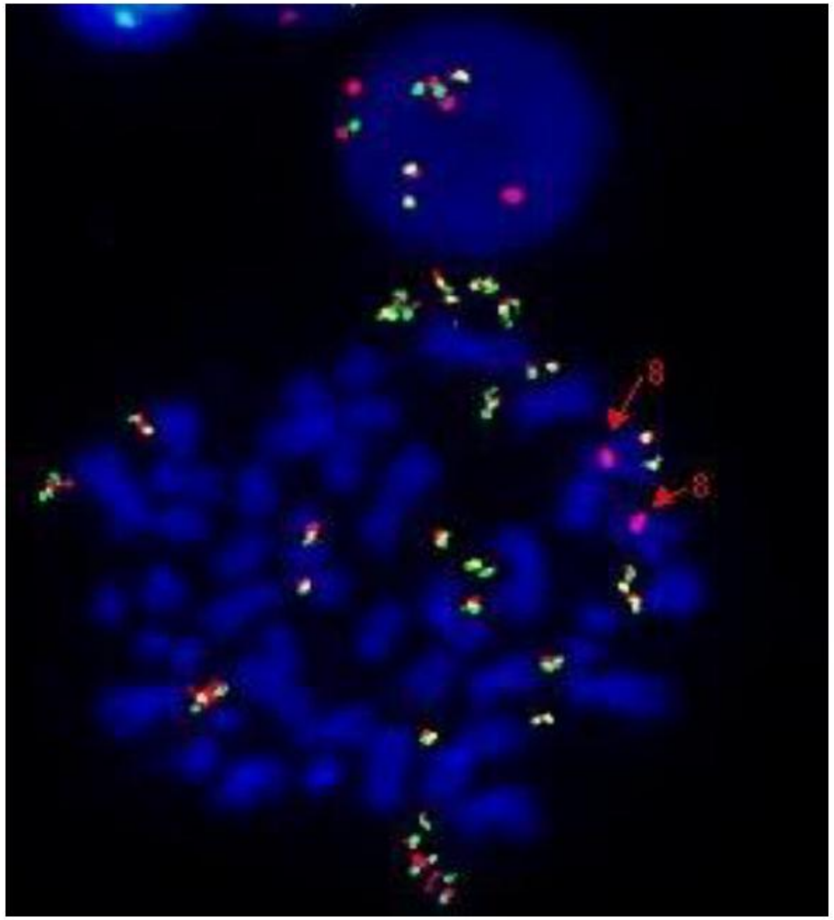
- Double minutes are extrachromosomal fragments of DNA containing an amplified oncogene (Multiple/ hundreds of copies of the oncogene)
- Seen in aggressive tumors where the amplified region includes an oncogene
- E.g.: EGFR is often amplified as double minute chromosomes (painted in red) in advanced gliomas – visualized with FISH probes

Gene Amplification of an oncogene: Double minutes



From G-band karyotype

Results in production of excessive amounts of the oncoprotein and a sustained stimulus to divide



Visualized with FISH probes

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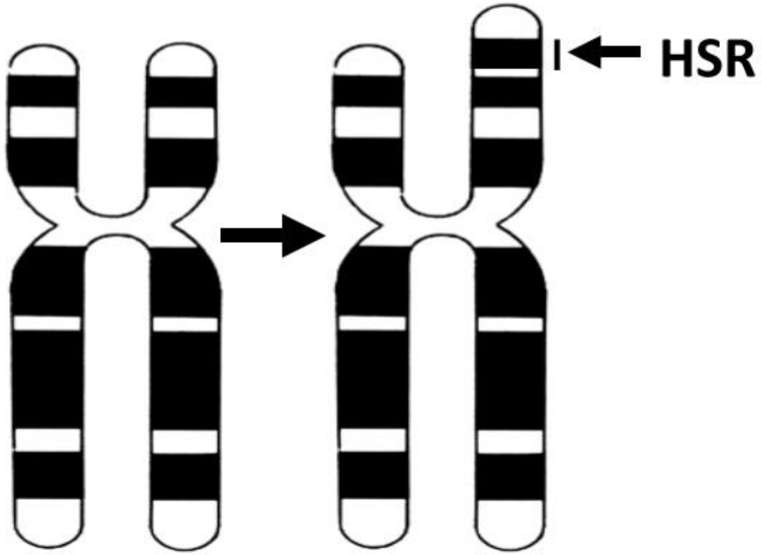
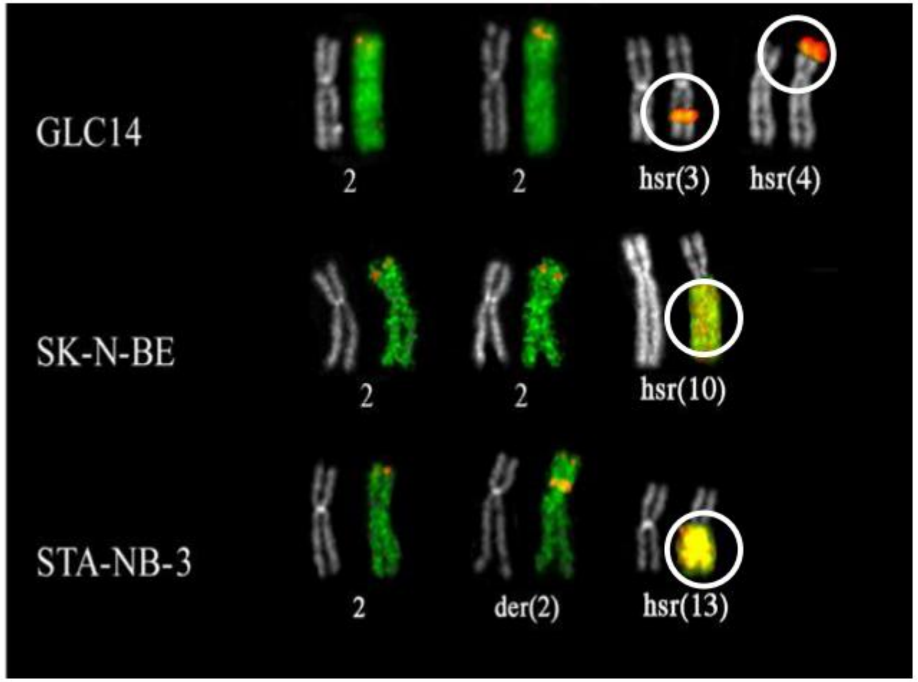
Oncogene Amplification: Homogenously staining regions

Abnormal Homogeneously Staining Regions (hsrs) contain amplified oncogenes (Multiple/ hundreds of copies of the oncogene)

The amplified oncogene is attached to the chromosome

e.g. N-MYC amplification in neuroblastomas (NB)





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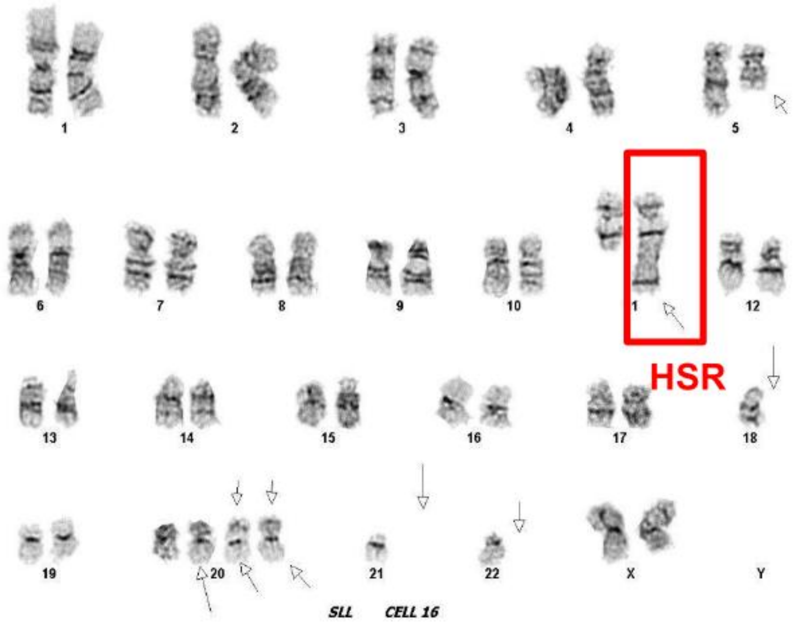
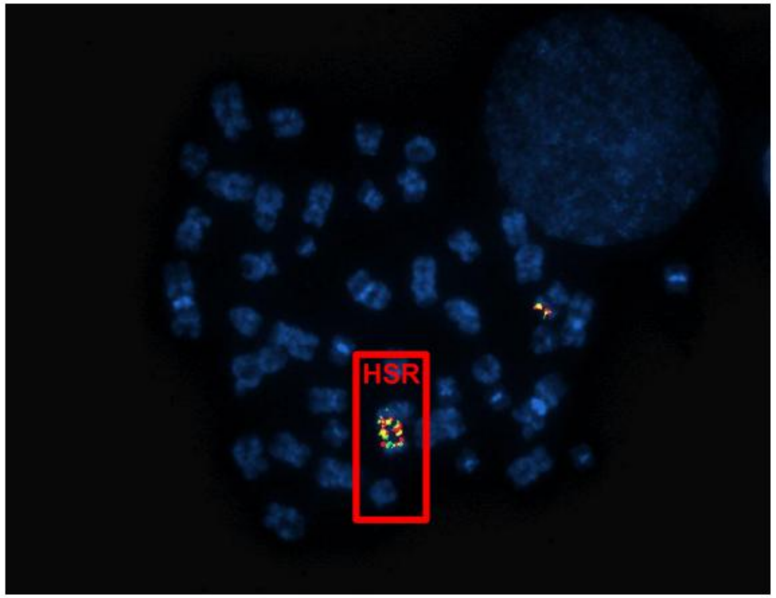
Gene Amplification of an oncogene: Homogenously staining regions



From G-band karyotype

Visualized with FISH probes

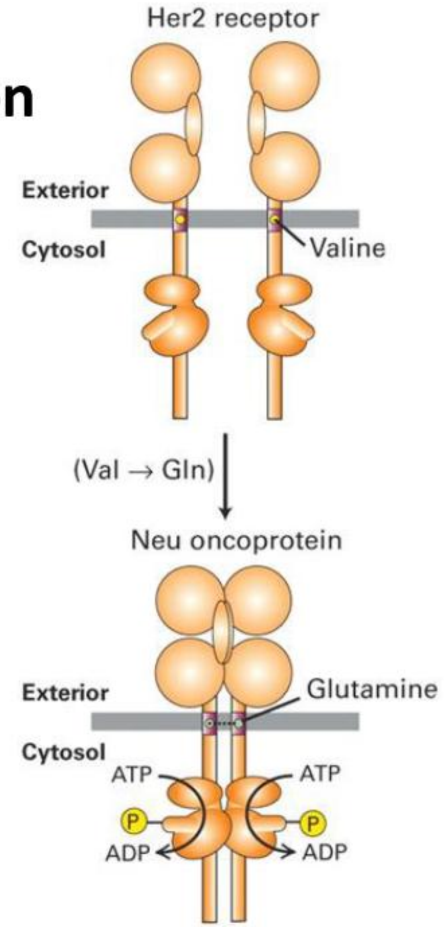


Double minutes are extrachromosomal fragments of DNA Whereas, hsrs are amplified oncogenes attached to the chromosome

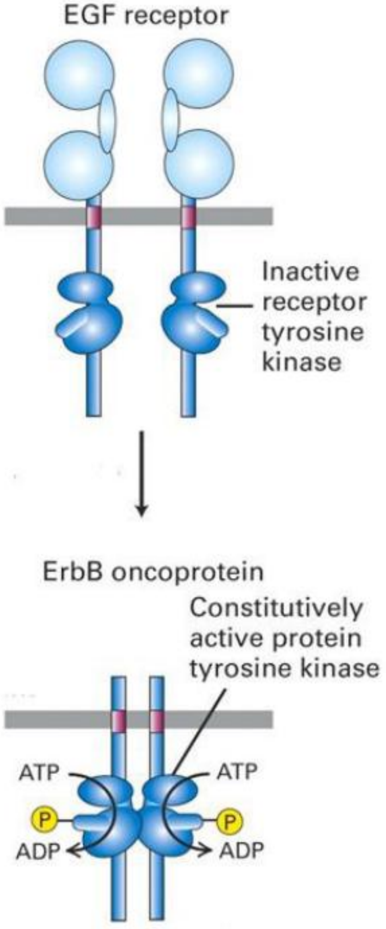
Oncogene amplification as Double minutes or hsrs result in overproduction of the oncoprotein resulting in an increased stimulation for the cell to divide

Truncations or point mutations can render a Growth Factor Receptor constitutively active

Point mutation



Truncation/ deletion



The abnormal protein is active even in the absence of a signal (constitutively active)

Point Mutation causing Oncogenic Activation of Growth Factor Receptor

- *ERBB1* is a gene that encodes the epidermal growth factor receptor (EGFR).

- Point mutations in *ERBB1* results in constitutive activation of the EGFR tyrosine kinase. This is seen in a subset of lung adenocarcinomas.

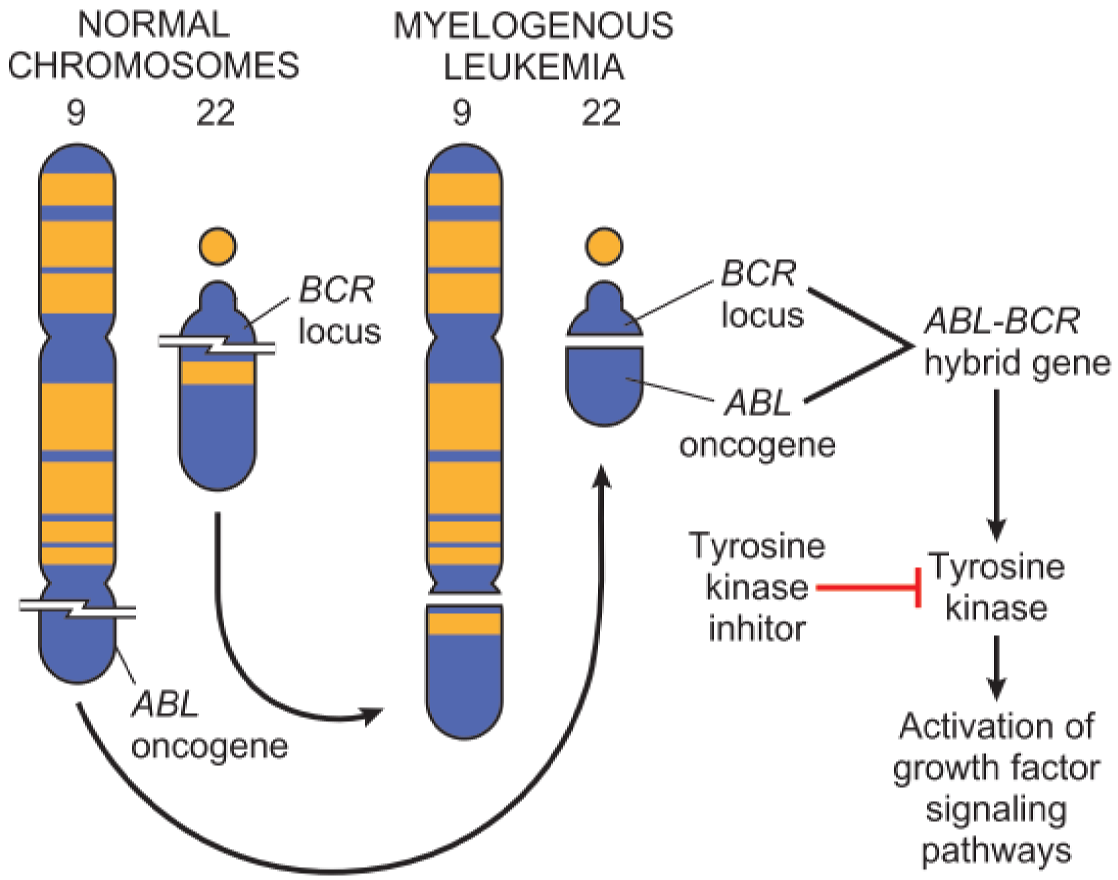
- Inhibitors of *ERBB1* are used therapeutic in lung adenocarcinomas.

Case 2 – Chronic Myelogenous Leukemia

- A 53-year-old man comes to physician because of fatigue, poor appetite, and unintentional weight loss. Physical examination shows conjunctival pallor. Lab findings show a white blood count of 24,700 per mL (normal is 4,000 – 7,000 per mL), hemoglobin of 9.9 g/dL (normal is 13.5 g/dL – 17.5 g/dL), and a platelet count of 557,000 per mL (normal is 150,000 – 500,000 per mL). Cytogenetic analysis with fluorescent *in situ* hybridization (FISH) confirmed a 9:22 translocation (Philadelphia chromosome), and quantitative reverse transcription polymerase chain reaction (qRT-PCR) detected the expression of the *BCR-ABL* gene confirming the diagnosis of Chronic Myelogenous Leukemia. He is put on Imatinib as first line therapy for CML.

How did a Chromosomal Translocation lead to Chronic Myelogenous Leukemia?

- In CML there is a chromosomal translocation that translocate the gene *ABL* from its normal position on chromosome 9 to a new location on chromosome 22 next to the gene *BCR*, activating *ABL*.
- This results in fusion gene *BCR-ABL* that encodes a constitutively active, nonreceptor tyrosine kinase. This tyrosine kinase is not bound to the cell membrane, but it activates the same signaling pathway of a receptor tyrosine kinase.



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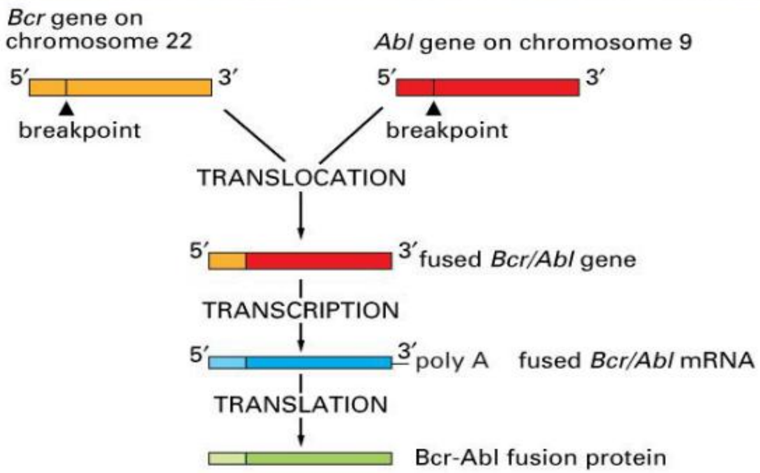
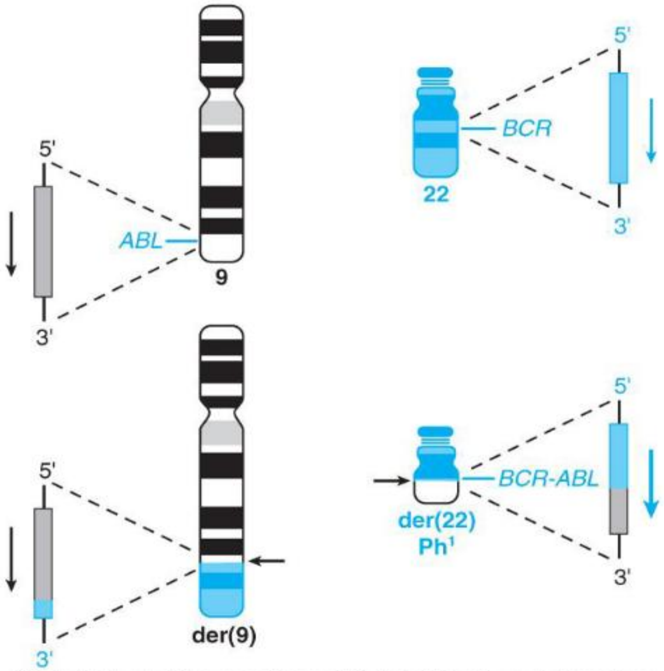
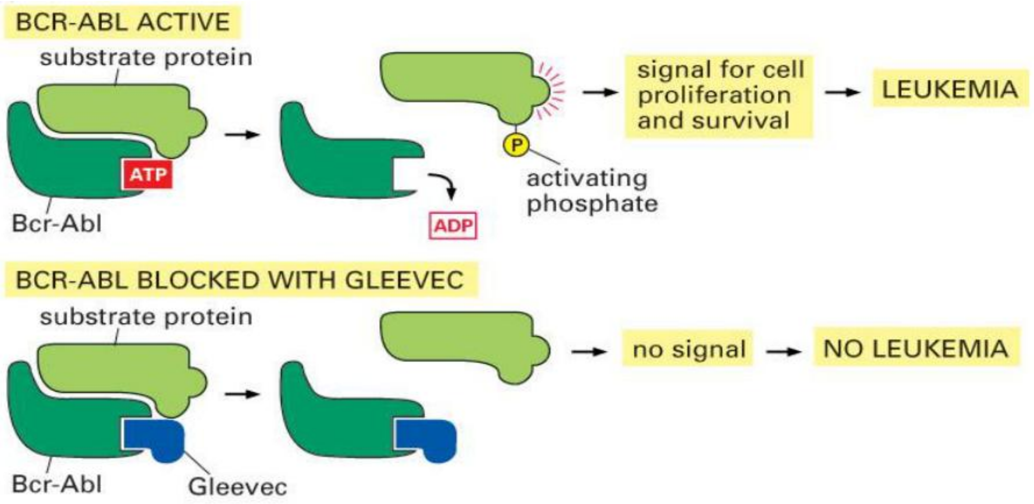
The Philadelphia Chromosome Translocation, t(9;22)(q34;q11)





Targeted Molecular Therapy in Chronic Myelogenous Leukemia

- Most CML tumor cells require the *BCR-ABL* signaling pathway to survive and proliferate. Thus, inhibiting its activity is a successful form of therapy.
- Imatinib mesylate (Gleevec) is a *BCR-ABL* tyrosine kinase inhibitor that is highly effective in CML as it suppresses the tumor proliferation

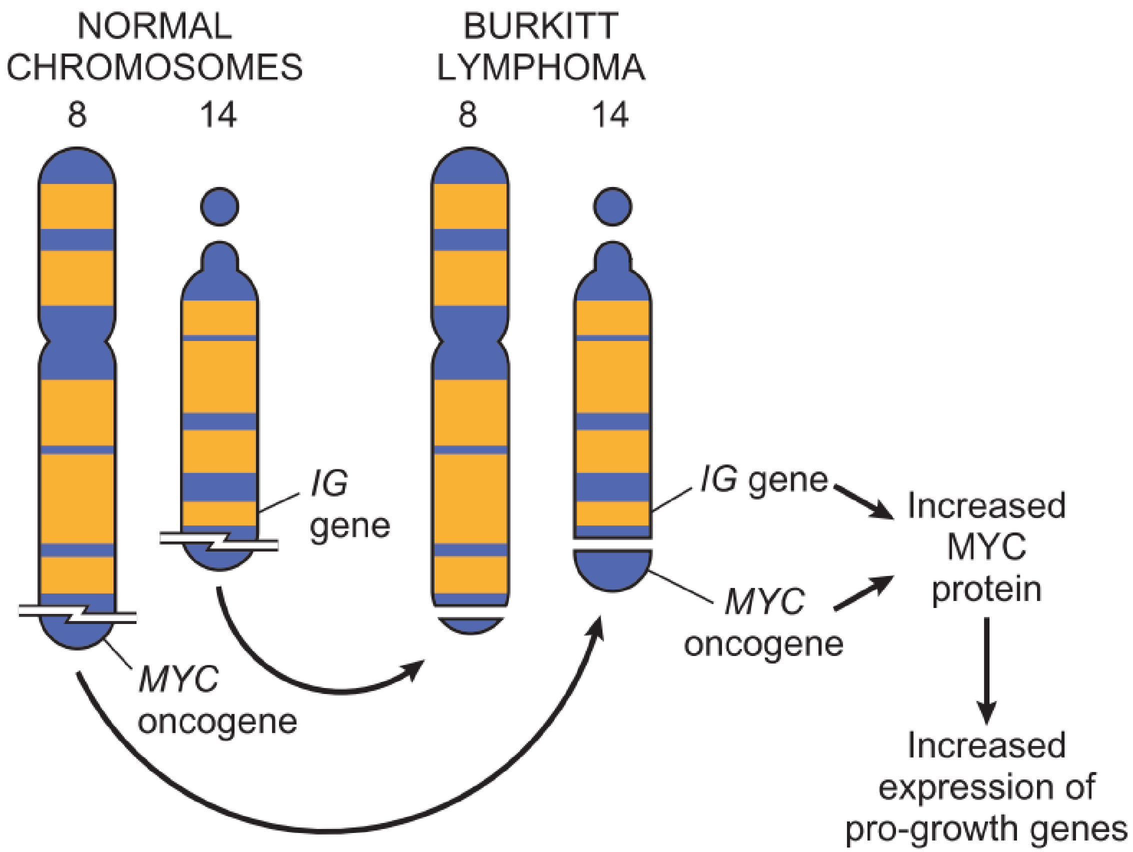


Case 3 – Burkitt's Lymphoma

- A 7-year-old boy is brought to the physician because of abdominal pain, weakness, and intermittent fever for the past 6 months. Physical examination shows generalized lymphadenopathy and splenomegaly. Laboratory findings show a WBC count of 2,700 per mL, Hemoglobin of 10.7 g/dL, and platelets of 178,000 per mL. Bone marrow aspirate shows is hypercellular with 30% lymphoid blasts. Flow cytometric analysis shows a B cell population. The chromosome analysis shows an 8:14 translocation. This confirmed the diagnosis Burkitt's lymphoma.

How did a Chromosomal Translocation lead to Burkitt's Lymphoma?

- Burkitt's Lymphoma results from an 8:14 translocation that results in *C-MYC* increased expression because it is now under the regulation of IgH (Active Promoter)
- *MYC* proto-oncogene is a transcription factor that activates the expression of many genes that are involved in cell growth. Thus it plays a role in G1/S phase transition.



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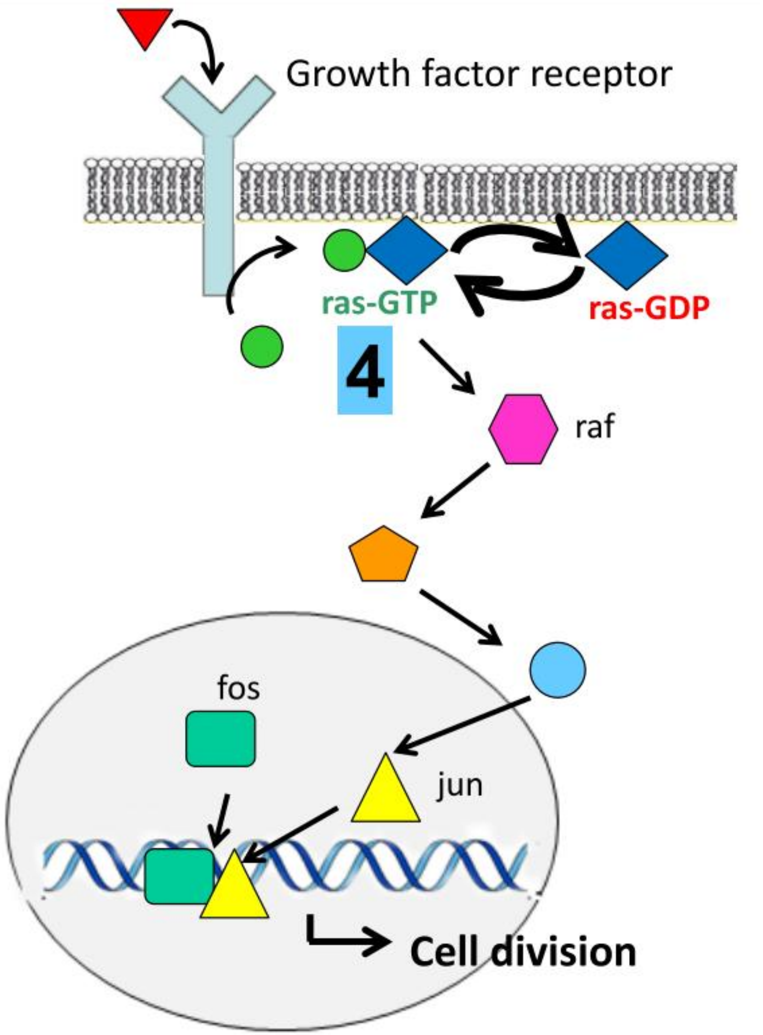
Activation of the Ras proto-oncogene

- Ras is a GTPase involved in the major cell proliferative pathway (MAP-Kinase)
- Ras activated by binding GTP
- It initiates a phosphorylation cascade that activates cellular proliferation
- It is quickly inactivated by intrinsic GTPase activity (GTP→GDP)
- A constant stimulation is required to continue to grow



Activation of the Ras proto-oncogene

- Oncogenesis depends on Ras hyper-activity (overactivity)
- Ras-GTP activates Growth pathway
- Ras-GDP inactive (No growth)
- Many mutations inhibit GTPase activity = Ras becomes constitutively active (Ras is active even in the absence of a growth factor)
- Single point mutants at Gly12 or Gln61 have been isolated from tumor cells that result in constitutive activation of ras